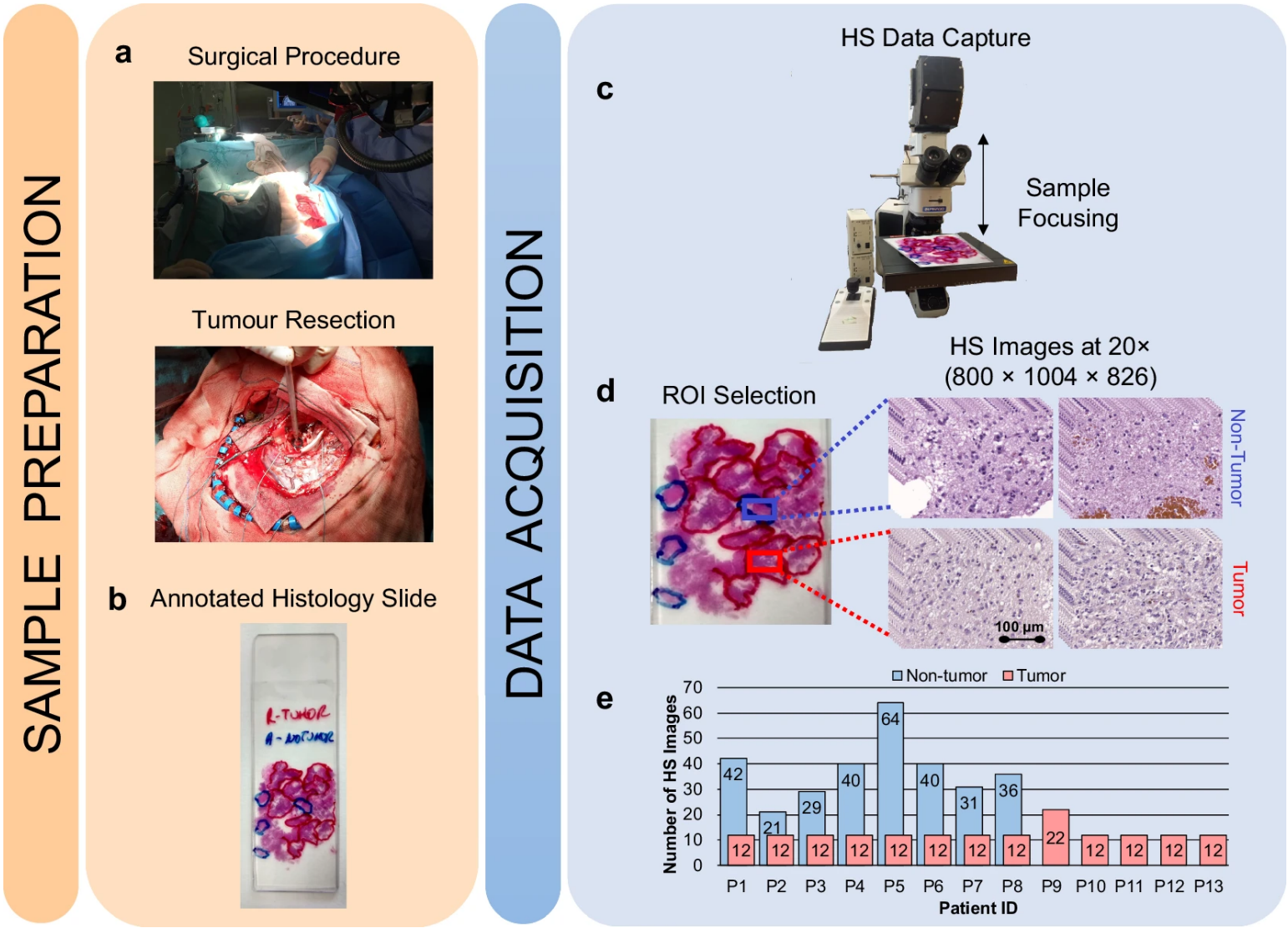
Histology HSI GB | Hyperspectral Histological Images for Diagnosis of Human Glioblastoma

DOI: 10.7937/z1k6-vd17 | Data Citation Required | 5k Views | IMAGE COLLECTION

Location	Species	Subjects	Data Types	Cancer Types	Size	External Resources
Brain	Human	13	Histopathology, Classification	Glioblastoma	582GB	Software/Source Code



From: Histological Hyperspectral Glioblastoma Dataset (HistologyHSI-GB) - https://www.nature.com/articles/s41597-024-03510-x

Hyperspectral imaging technology combines the main features of two existing technologies: conventional imaging and spectroscopy. Thus, hyperspectral cameras make it possible to analyze, at the same time and in a non-contact way, the morphological features and chemical composition of the objects captured. The information provided by hyperspectral imaging can be used to detect patterns, cells, or biomarkers to identify diseases. There are different alternatives for processing them and there is a lack of publicly available datasets of medical hyperspectral images. To the best of our knowledge, this is the first open access dataset containing histological hyperspectral images of glioblastoma brain tumors, which can be set as a benchmark for researchers to compare their approaches.

This dataset includes 13 subjects. Each subject has a single histological slide with multiple hyperspectral images captured from each slide where deemed relevant by the pathologists (this

number varies for each slide). The database is composed of 469 annotated hyperspectral images from 13 histological slides (482 total images), having a spatial dimension of 800 × 1004 pixels and a spectral dimension of 826 spectral channels. The format of the hyperspectral images is ENVI, the standard format for the storage of hyperspectral images. The ENVI format consists of a flat-binary raster file which may or may not have a file extension, accompanied by an ASCII header file (denoted as *.hdr). The data are stored in band-interleaved-by-line format. In addition, dark and white references were captured to perform a calibration of the raw image, which is a standard procedure in hyperspectral image processing.

The slides were stained with hematoxylin and eosin and captured using a custom hyperspectral microscopic system at 20× magnification. The ground-truth annotation for this dataset is the diagnosis of the slides (*tumor* _T_ or *not tumor* _NT_) performed by skilled histopathologists after the visual examination of the stained slides, according to the World Health Organization classification of tumors of the nervous system. As far as we are concerned, there are no commercial hyperspectral whole slide scanners. Also, the availability of hyperspectral microscopes is still limited in the market.

The microscope is an Olympus BX-53 (Olympus, Tokyo, Japan). The hyperspectral camera is a Hyperspec® VNIR A-Series from HeadWall Photonics (Fitchburg, MA, USA), which is based on an imaging spectrometer coupled to a charge-coupled device sensor, the Adimec-1000m (Adimec, Eindhoven, Netherlands). This hyperspectral system works in the visual and near-infrared spectral range from 400 to 1000 nm with a spectral resolution of 2.8 nm, sampling 826 spectral channels, and 1004 spatial pixels. The push-broom camera performs a spatial scanning to acquire a hyperspectral cube with a mechanical stage (SCAN, Märzhäuser, Wetzlar, Germany) attached to the microscope, which provides an accurate movement of the slides. The objective lenses are from the LMPLFLN family (Olympus, Tokyo, Japan), optimized for infrared observations.

More information about the dataset can be found in this manuscript.

Data Access

Version 1: Updated 2024/05/24

Title	Data Type	Format	Access Points	Subjects	St
Tissue Slide Images	Histopathology, Classification	PNG and ENVI	DOWNLOAD (582GB) SEARCH Download requires IBM-Aspera-Connect plugin	13	

RELATED DATASETS

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External Resources

The following external resources have been made available by the data submitters. These are not hosted or supported by TCIA, but may be useful to researchers utilizing this collection.

- A tutorial on how to read and display the hyperspectral data using Python or MATLAB is available at https://github.com/HIRIS-Lab/HistologyHSI-GB.
- The authors also recommend GLIMPS (https://www.rese-apps.com/software/download/form.html) for the visualization of ENVI images.

Citations & Data Usage Policy

Data Citation Required: Users must abide by the TCIA Data Usage Policy and Restrictions. Attribution must include the following citation, including the Digital Object Identifier:

Data Citation

Ortega, S., Fabelo, H., Quintana-Quintana, L., Leon, R., Plaza, M.d.l.L., Camacho, R., & Callico, G. M. (2024). Hyperspectral Histological Images for Diagnosis of Human Glioblastoma (HistologyHSI-GB) (Version 1) [Data set]. The Cancer Imaging Archive. https://doi.org/10.7937/Z1K6-VD17

Acknowledgements

Related Publications

ABOUT THE CANCER IMAGING ARCHIVE (TCIA)

TCIA is a service which de-identifies and hosts a large archive of medical images of cancer accessible for public download. The data are organized as "collections"; typically patients' imaging related by a common disease (e.g. lung cancer), image modality or type (MRI, CT, digital

ABOUT THE CANCER IMAGING PROGRAM (CIP)

The Cancer Imaging Program (CIP) is a part of the Division of Cancer Treatment and Diagnosis (DCTD) of the National Cancer Institute. For complete information about the Cancer Imaging Program, please see the Cancer Imaging Program Website.

HOSTING TCIA AT UAMS

In October 2015 Dr. Prior and the core TCIA team relocated from Washington University to the Department of Biomedical Informatics at the University of Arkansas for Medical Sciences. The archive continues provides high quality, high value image collections to cancer researchers

ABOUT FNLCR

The Frederick National Laboratory for Cancer Research (FNLCR) is a Federally Funded Research and Development Center dedicated to the application of biomedical science and technology to improve human health. FNLCR scientists conduct basic, translational and applied research, create new

histopathology, etc) or research focus. DICOM is the primary file format used by TCIA for radiology imaging. Supporting data related to the images such as patient outcomes, treatment details, genomics and expert analyses are also provided when available.

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technologies, and collaborate with government, industry and academic colleagues to support the National Cancer Institute and other Institutes of the National Institutes of Health.

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